University of Texas at Dallas

Department of Electrical Engineering

EEDG/CE 6306 - Application Specific Integrated Circuit Design

Midterm Project

Submission:

(a) Verilog block- and top-level programs & testbenches of MSDAP (35pts)

(b) All input and output files used for unit testing at block-level (e.g. rj, coeff and input data file) (20pts)

(c) Top-level Output file data2.out (10pts)

(d) Synopsys Design Compiler output files and reports showing that the top-level verilog programs are all synthesizable (10pts)

(e) Written Report (PDF) (20pts)

(f) Presentation Slides (PDF) (5pts)

Related files are posted on eLearning, which includes

- **(a) data1.in & data1.out**
- **(b) data2.in**

Submit onto eLearning a zip file (team): <First name #1>_<Last name #1>_<FIrst name #2>_<Last name#2>_MidtermProject.zip

* If you have any question about the homework, you may contact the TA

Purpose: Synthesizable Verilog to create the MSDAP which includes block- and top-level (Datapath and Controller blocks) and their testbenches (block- and top-level).

In this assignment we will design the block-level and top-level verilog inside of the MSDAP.

(1) Write Verilog programs to implement each datapath block according to your block specifications. Create block-level testbenches to test each

datapath block. Use the midterm project inputs posted on eLearning to test your code.

The memory blocks will be behavioral for now. We will introduce memory macros in future assignments.

- Include in your written report screenshots of the simulation result (waveform) from ModelSim.

(2) Write Verilog programs to implement all of the Controllers/FSM's in the MSDAP. All the states and state transitions must be shown in the report through a state transition graph. You may choose to implement the FSM which is shown below that was discussed in class or design your own. Create testbenches and test all of the states and transitions in each controller.

- Include in your written report screenshots of the simulation result (waveform) from ModelSim.

(3) Write a top-level Verilog program to integrate all of the datapath and control blocks in the MSDAP. Create an external controller testbench (Figure 1), that takes in the rj/coeff/data input files (data1.in, data2.in) and outputs data (data1.out, data2.out).

- Generate the input signals whose format matches with the Spec.
- Collect outputs from the MSDAP chip and generate the output file.
- Include in your written report screenshots of the simulation result (waveform) from ModelSim.

Note: Test files can be found on eLearning. In the midterm project, all R_js, coefficients and data are in the same file. Please parse them within your testbench (e.g. don't create new input files). There are two sets of input, data1.in and data2.in. data1.in is with the reference output file data1.out for the test purpose. And the other one, data2.in is used to generate the output of the submission. The occurrences of reset are marked in the comment section of the input file.

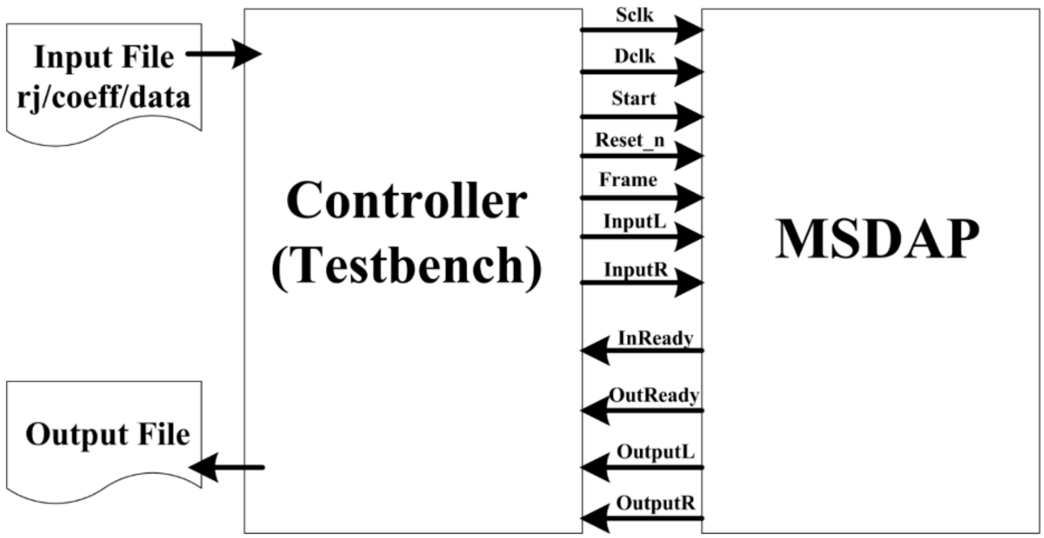

Figure 1. Top-level testbench

(4) Synthesize your top-level MSDAP design and all blocks using Synopsys Design Compiler. Include the output log file that shows that the top-level has successfully synthesized.

(5) Create a written report and a 15-min presentation - include a PDF of the report and the presentation files in your submission

(5.1) Top-level and Block-level architecture diagrams, description and decisions made. Talk about the performance of your MSDAP.

(5.2) Include any top-level and block-level controllers, state transition diagrams.

(5.3) Include top-level and block-level waveforms\figures\tables to justify your design.

(5.4) Did you need to modify your block- or top-level specifications? Discuss any modifications and why.

(5.5) Talk about any challenges you faced during the project.

(5.6) Discuss your Generative AI strategy and any observations / key learnings from this strategy.

(5.7) Reserve a time slot for your 15-min group presentation on this Google Sheet.